



Society of Petroleum Engineers

SPE-226609-MS

Novel Production Enhancement Approach – Multi-Stage Acid Frac to Improve Gas Production and Maximize Reserve of the Tight Dolomite Layers in the Bahrain Field

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This paper was prepared for presentation at the SPE International Hydraulic Fracturing Technology Conference and Exhibition held in Muscat, Sultanate of Oman, 23 - 25 September 2025.

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Abstract

The primary objective of this comprehensive study is to present the detailed process and highly successful outcomes of the first hydraulic fracturing (frac) job carried out in one of the gas reservoirs in Awali Field. This endeavor is specifically aimed at evaluating the untapped potential of the tight intervals that lie within the four distinct layers of the subject reservoir. These tight layers have previously been overlooked in conventional completion strategies. By concentrating on these underdeveloped zones, the study aspires to reveal and illustrate the significant production capacity that has remained unexploited. Through this innovative approach, the study aims to provide a replicable framework that can enhance overall gas production and resource utilization in similar

The traditional completion strategy involves drilling and completing of wells by perforating the four high permeability reservoir layers. This is followed by an acid wash of the perforation intervals and producing the well as a commingled single completion. In order to evaluate the latent potential of the tight layers, a new seven-inch well was drilled, and a multi-stage acid fracturing (frac) technique was carried out, specifically targeting selected tight intervals within the four high permeability layers. Besides testing each layer individually, surveillance data such as production logs and pressure buildup data are gathered to evaluate the performance of each layer. Subsequently, a commingled completion was installed to assess the combined productivity of these tight intervals.

Use of multi-stage acid frac targeting the tight intervals in the new well, showed exceptionally promising results. Each of the four distinct layers exhibited high productivity rates when tested separately. Production logging data indicated higher inflow contribution from two tight layers. In addition to higher productivity of tight intervals, pressure buildup results show pressures higher than average field pressure suggesting additional untapped gas potential. Production decline rate is also lower than the wells with normal completion well signifying additional reserves. The results of this study indicate that this innovative approach effectively harnesses the potential of previously overlooked tight layers en route to maximizing productivity and recoverable reserves.

The introduction of a multi-stage acid frac technique specifically targeting the tight intervals represents a novel and groundbreaking approach in the field of gas production. The successful demonstration of this technique provides a robust and replicable framework and highlights the potential for substantial enhancement in gas production not only from the subject reservoir but also from similar reservoirs thereby contributing to the overall gas resource management. The findings of this study underscore the importance of innovative approaches in the sustainable and efficient management of energy resources in the Bahrain Field.

Introduction

Since its discovery, the Awal field—situated above the large anticline forming Bahrain Island—has been a principal gas producer in Bahrain, targeting Cretaceous and Jurassic carbonates. Gas production from the main limestone reservoir started in 1969 with an increasing trend to fulfil the Kingdom's requirements. It is a common practice to target high porosity and high permeability layers to achieve maximum productivity from drilled wells. With higher recovery factor, targeting untapped gas reserves reside in tight dolomite layers characterized by low permeability ($< 1\text{--}10\text{mD}$) and limited natural fractures which is allocated in tighter intervals becomes a target to achieve higher production rates and increase reserves and recovery factor. The drive to maximize recovery from these tight formations is critical to meeting future domestic energy needs and extending field life. These tight intervals represent up to 70% of the reservoir rock with less than 5% in terms of production rate. Fig. 1 displays a typical Khuff type log. Fig. 2 shows the main flow component in Khuff gas reservoir.

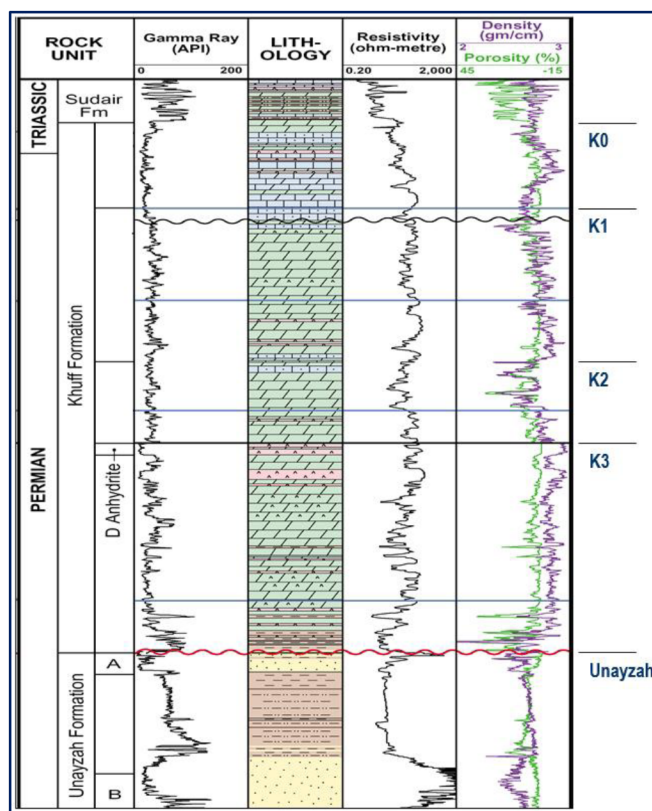


Figure 1—Khuff reservoir type log

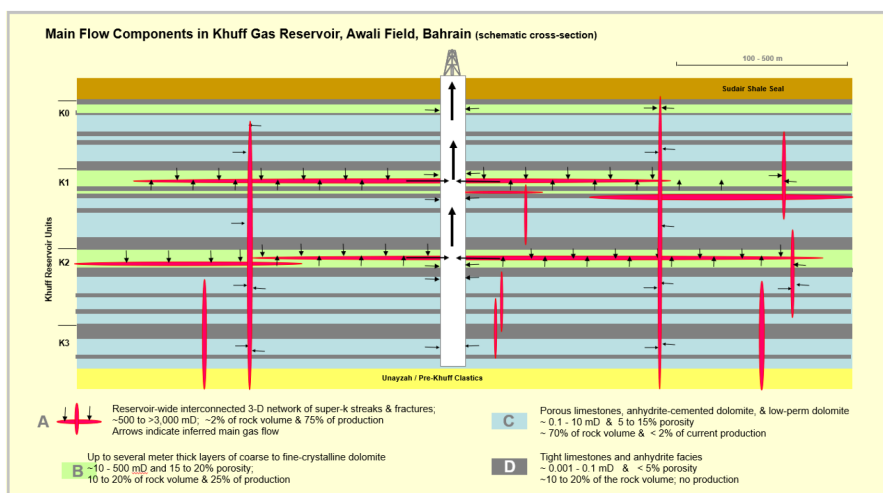


Figure 2—Main flow component in Khuff gas reservoir in Bahrain

Khuff Reservoir Rock and Fluid Characteristics

Table 1—Rock and Fluid Properties of Khuff Reservoir

Parameter	Value
Permeability	K0: 2.4 mD; K1:30 mD; K2:20 med; K3: 0.02 mD
Porosity	K0: 12%; K1:10%; K2:10%; K3:6%
Net pay thickness	K0: 20 ft.; K1:55 ft.; K2:97 ft.; K3: 1ft.
Reservoir pressure (initial)	6,330 psi @ -9600 TVD
Reservoir pressure (current)	1,650 psi (Khuff)
Reservoir temperature	270°F
GOR	120,000 – 185,000 scf/STB
Initial gas saturation	K0: 84%; K1:8 9%; K2: 91%; K3: 45%
Dew point pressure	3,700 – 4,948 psi (Avg. 4,450 psi)
Gas viscosity (reservoir conditions)	0.03399 cP
Initial CGR	5 – 8.3 stb/MMscf
Gas formation volume factor	0.65524 rb/Mscf @ 6,350 psi
MW	20.1 – 20.4 lb-lbmol

Khuff Primary Production Approach

Khuff gas reservoir consist of 4 dolomite layers named K0, K1, K2, and K3 (Fig. 3). These layers have different rock and formation properties with some intervals exceeding 1 Darcy in permeability. Fig. 4 shows a cross section of the reservoir representing variation in porosity distribution vertically in different location. To optimize completion strategy, wells were completed as multi-layer targeting best intervals in terms of porosity and permeability. Targeting these intervals was requiring a simple acid wash to remove any possible skin created by drilling fluid and put these wells on production with several wells being choked based on surface facilities limitations. With higher recovery factor and depletion in reservoir pressure, matrix acid stimulation became more frequent and essential to maximize well's productivity. This practice was not focusing on tight intervals in each layer as tight intervals cannot flow through the same completion with the availability of high permeability intervals on production.

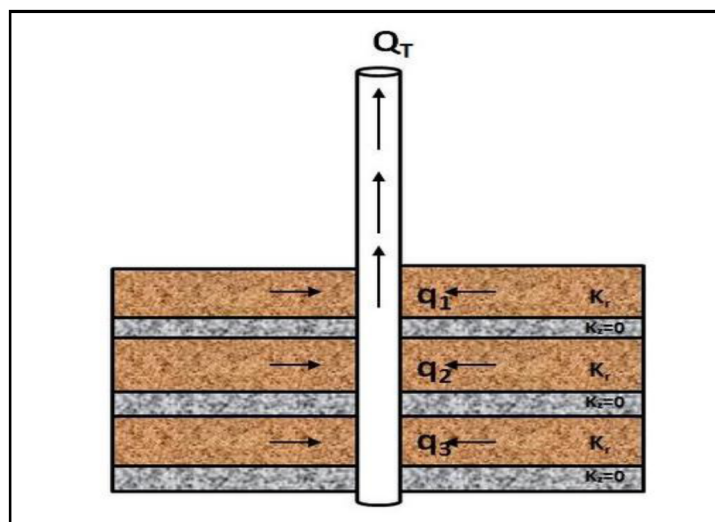


Figure 3—Primary production methodology

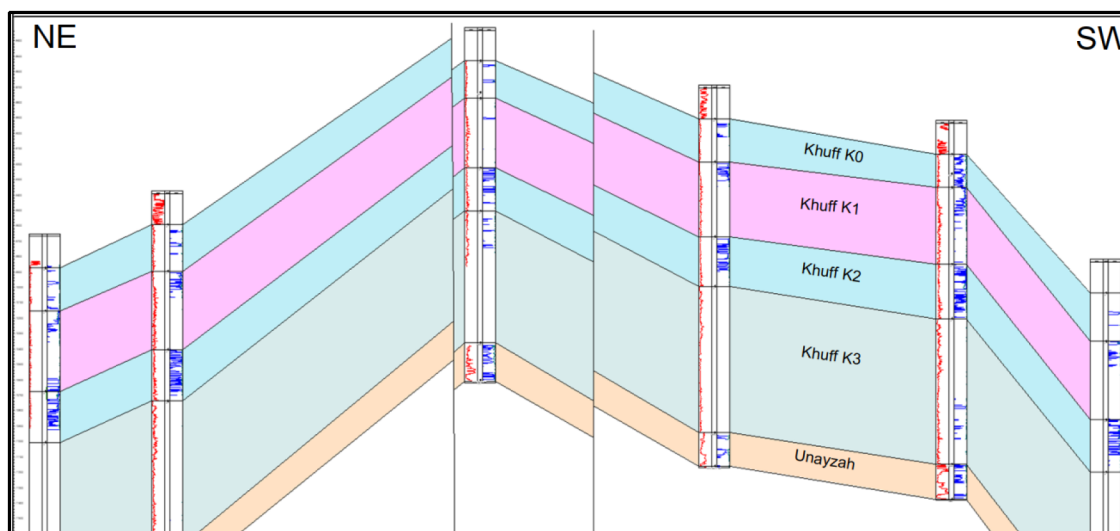


Figure 4—Gas reservoir cross section showing main porosity intervals.

Core data analysis provided a correlation between porosity and permeability, which indicated the availability of areas with moderate porosity with low permeability as displayed in Fig. 5.

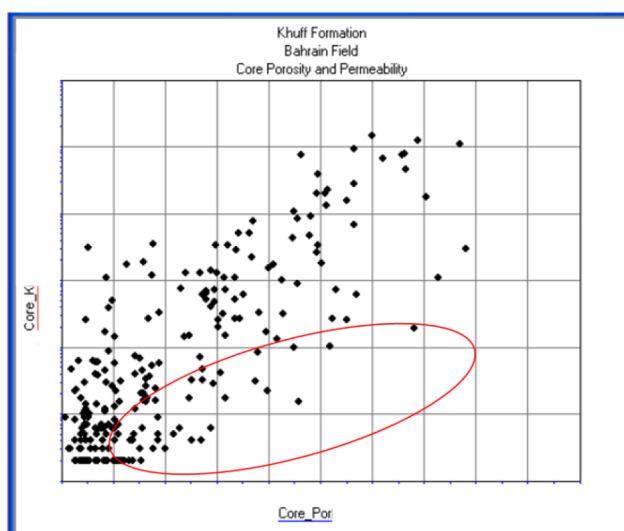


Figure 5—Core porosity and permeability relationship

Acid fracturing

Acid fracturing is one of the most effective measures for carbonate stimulation, and become the go-to stimulation method in tight carbonates where conventional hydraulic fracturing struggles due to small pore throat sizes. Historic SPE studies in China and Poland confirm that acid frac—notably gelled, emulsified, or Syracuse-formulated acids—effectively etches fracture networks, enhancing conductivity without the need for proppant placement (Nasr El Din et al., 2003; Xiong et al., 2017). This is also supported when natural fractures are present in the formation. A study reviewing Mishrif ("Rubble") formation using fracture imaging identified domains with varying natural fracture networks. The study concluded that wells in high-fracture domains outperformed others, underscoring the importance of detailed fracture mapping for frac planning (Stearns et al., 2014). Additionally, tight dolomite layers are subject to water-lock damage due to water saturation from drilling and workover fluids. Chemical drainage methods—especially acid fragmentation—help mitigate this by improving gas relative permeability (Bugti et al., 2010; Patidar et al., 2022)

The development of unconventional reservoirs, particularly tight dolomite formations, presents significant challenges due to their low permeability and complex geological characteristics. Multi-Stage Acid Fracturing (MSAF) has emerged as a promising stimulation technique to enhance gas production and optimize hydrocarbon recovery in such reservoirs. Unlike conventional hydraulic fracturing, MSAF leverages acid-based fluids to dissolve carbonate minerals, creating highly conductive pathways that improve reservoir connectivity and fluid flow.

Recent studies highlight the effectiveness of MSAF in stimulating deep carbonate formations, particularly in regions with ultra-deep subsalt dolomite reservoirs. The process involves sequential acid injection stages, which enhance fracture conductivity and maximize the stimulated reservoir volume. Research indicates that acid worm holing plays a crucial role in improving fracture permeability, allowing for better hydrocarbon extraction, and suggests a two-phase injection: high-rate for fracture propagation and low-rate for maximizing etched conductivity considering temperature-sensitive reactivity and avoiding excessive acid loss (Zhu et al., 2023; Wang et al., 2018). Additionally, optimized acid systems and fluid combinations have been shown to significantly increase the efficiency of acid fracturing treatments.

One of the key advantages of MSAF is its ability to enhance fracture conductivity through controlled acid etching. Unlike proppant fracturing, which relies on mechanical support to maintain fracture width, acid fracturing modifies the rock structure, creating long-lasting conductive channels. Studies have demonstrated

that multi-stage alternating injection of pad and acid fluids can lead to narrower and longer acid-etched channels, improving overall reservoir stimulation.

Field applications of MSAF, such as those conducted in Middle Eastern carbonate reservoirs, demonstrate its potential in rejuvenating mature reservoirs and increasing gas production. By integrating advanced acid fracturing techniques with tailored fluid systems, operators can achieve higher recovery rates and maximize reserves, ensuring sustainable production from tight dolomite formations.

This paper explores the principles, mechanisms, and case studies of MSAF, providing insights into its application for improving gas production and maximizing reserves in tight dolomite layers.

Discussion of Previous Studies on Acid Fracturing, Carbonate Reservoirs, and Reservoir Stimulation Techniques

Acid fracturing is a widely used stimulation technique in carbonate reservoirs to enhance permeability and improve hydrocarbon recovery. Previous studies have explored various aspects of acid fracturing, including its effectiveness in deep and ultra-deep carbonate reservoirs, challenges related to acid leak-off, and optimization of acid systems. Research has also focused on thickened acid fracturing, which improves acid retention and etching efficiency in carbonate formations (Al Jawder et al. 2022). Additionally, case studies have compared acid fracturing with propped fracturing in tight-gas carbonate reservoirs, highlighting the advantages and limitations of each approach.

Comparison of Conventional Fracturing vs. Multistage Acid Fracturing in Vertical Well Applications

Conventional fracturing typically involves a single-stage treatment, which may not be sufficient for complex reservoirs with varying permeability. Multistage acid fracturing, on the other hand, allows for targeted stimulation of multiple zones within a vertical well, leading to improved reservoir contact and enhanced production. Studies have shown that multistage fracturing increases production rates and overall, well productivity by optimizing fracture placement and orientation (Li et al. 2006). Additionally, advancements in multistage fracturing technology have addressed challenges such as tool abrasion and frac string integrity, making it a viable option for vertical wells in low-permeability reservoirs.

Methodology

Overview of Acid Fracturing Process

The acid fracturing treatment in the newly drilled well was designed to stimulate production from four distinct carbonate formations—K-3, K-2, K-1, and K-0—using a multistage approach. Each stage targeted a single formation layer to ensure precise acid placement and maximize reservoir contact as shown in Fig. 6. The enhanced completion approach involves the following procedure:

1. The subject well was drilled and completed in Khuff (K3+K2+K1+K0).
2. After perforating each zone, the zone was acid fracked and tested.
3. After testing is completed, the zone was isolated by a plug and then another zone is perforated following the same procedure.

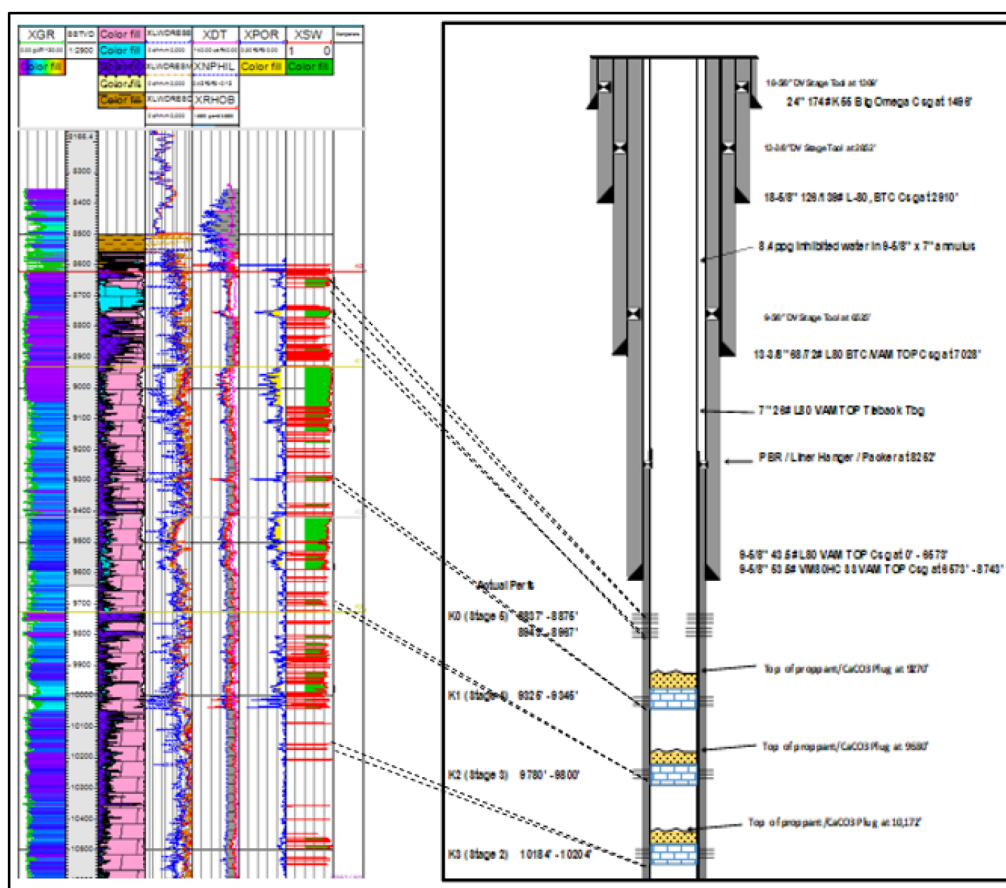


Figure 6—New completion strategy

The key process involved are.

- Pre-treatment injection tests to evaluate formation injectivity and estimate reservoir pressure.
- Main acid fracturing treatments using a combination of carbonate stimulation and emulsion acids, proppant-free pad fluids, and diverting agents.
- Post-treatment analysis (flow test, PTA) to assess treatment effectiveness and optimize future operations.

Fluid Systems and Injection Strategy

Fluid Composition. The fluid systems used across the stages included:

- 26% Carbonate Stimulation Acid: For primary rock dissolution and fracture etching.
- 26% Carbonate Emulsion Acid: To enhance acid penetration and reduce fluid loss.
- A high-viscosity linear gel used as a pad fluid to initiate and propagate fractures.
- A Diverting Agent: Deployed between acid stages to ensure zonal coverage and prevent premature acid breakthrough.
- Each fluid was quality-checked on-site for acid strength, iron content, and crosslinking properties. Rheology and stability tests confirmed fluid performance under downhole conditions.

Injection Strategy. The injection strategy was tailored for each formation based on injectivity tests and formation response:

- Injection tests were conducted prior to each main treatment to determine breakdown pressure and fracture gradient.
- Pumping schedules were designed with alternating acid and pad stages, interspersed with diverting agents to optimize acid distribution.
- Flush and overflush stages followed each treatment to displace fluids and clean the wellbore.

Fracture Design Considerations

Fracture design was based on the following:

- Formation depth and pressure gradient: Ranging from 0.24 to 0.43 psi/ft.
- Perforation strategy: Each stage had a dedicated perforation interval with controlled shot density and orientation.
- Treatment volumes and rates: Adjusted to maintain positive treating pressure and ensure acid penetration. Maximum slurry rates reached up to 67.5 bpm, with treating pressures up to 4,583 psi.

Experimental Setup and Field Implementation

Equipment and Monitoring. Surface equipment included high-pressure pumps, acid tanks, and real-time monitoring systems. INSITE for Stimulation software was used to monitor treating pressure, slurry rate, and bottom-hole pressure in real time. Blender additives were tracked to ensure consistent fluid properties throughout the job.

Field Execution. Each stage was executed with a pre-job safety and planning meeting. Fluids were batch-mixed and tested on-site. The acid reached the perforations as indicated by a sharp drop in treating pressure, confirming fracture initiation. LoGard was observed to increase treating pressure, validating its effectiveness in diverting acid to untreated zones.

Quality Assurance. Lab tests included viscosity, pH, and crosslinking time measurements. Field QA/QC ensured acid strength and cleanliness met design specifications. Post-job analysis confirmed successful stimulation with no HSE incidents.

Results, Observations, and Discussion

Production Testing

Following the multistage acid fracturing treatment, individual layer tests were conducted to evaluate the contribution of each formation. The results demonstrated that all four targeted carbonate layers responded positively to stimulation, with varying degrees of gas production and wellhead pressure (WHP). The highest productivity was observed in the deeper layers, with progressively lower rates in the shallower zones. This trend aligns with the expected pressure gradient and reservoir quality variations. The total production from all layers is comparable to a good Khuff well. The production performance of the well is shown in [Fig. 7](#) and the comparison with average khuff well and some of the good performing wells is presented in [Fig. 8](#).

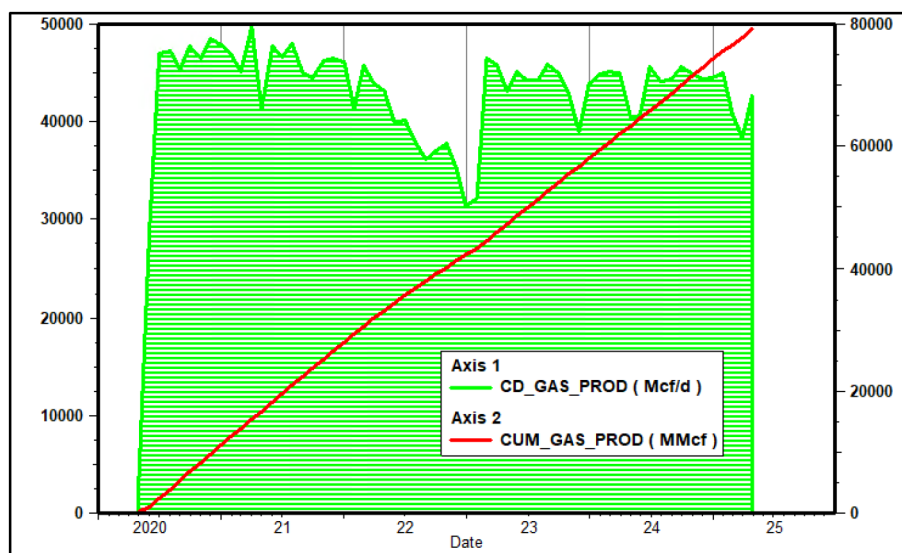


Figure 7—Production performance of the fracked well

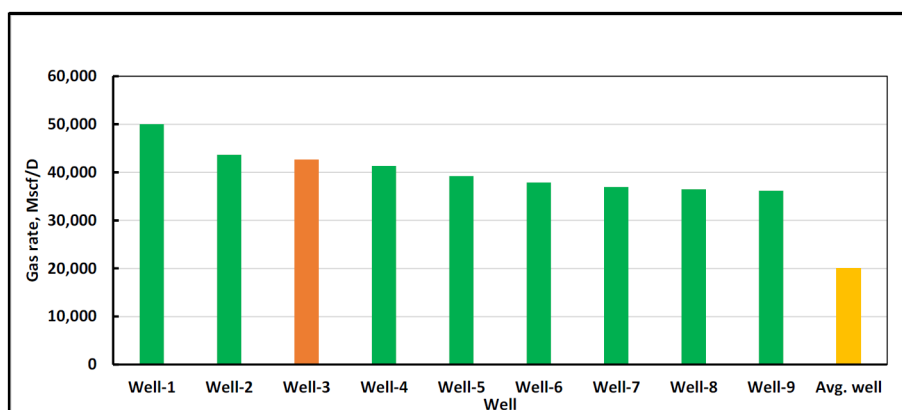


Figure 8—Comparison of production performance of the fracked well (Well-3) with average well and some of the good performing Khuff wells.

One upper zone, not part of the acid fracturing campaign, exhibited minimal flow and was later found to have sand accumulation at surface, indicating mechanical blockage or formation instability. This zone was subsequently isolated.

Pressure Transient Analysis (PTA)

PTA was successfully conducted for all four stimulated layers followed by isochronal tests. For brevity, the buildup and isochronal tests data are depicted in Fig. 9 for K3 reservoir only. The measured reservoir pressures ranged from approximately 2,198 psi in the shallowest zone to 2,559 psi in the deepest. These values are consistent with the pre-frac injection test estimates and confirm the presence of effective fracture conductivity and reservoir communication post-treatment. The log-log diagnosis shows half slope (1/2) indicating linear flow for K0 reservoir and negative mechanical skin factors for all the four zones suggest successful stimulation (Fig. 10). The fracture half-length for K0 reservoir based on linear flow analysis is 111 feet. The isochronal test data are used to match the test rates and estimate absolute open flow potential (AOFP) based both laminar-inertial-turbulent (LIT), pseudo-steady state and "C" & "n" analysis methods (Brar and Aziz, 1978; Lee and R. Wattenbarger, 1996). For brevity, only LIT analysis is depicted in Fig. 11. The matching of test rates with IPR plots lend credence on the methodology used and instill confidence on the analysis results.

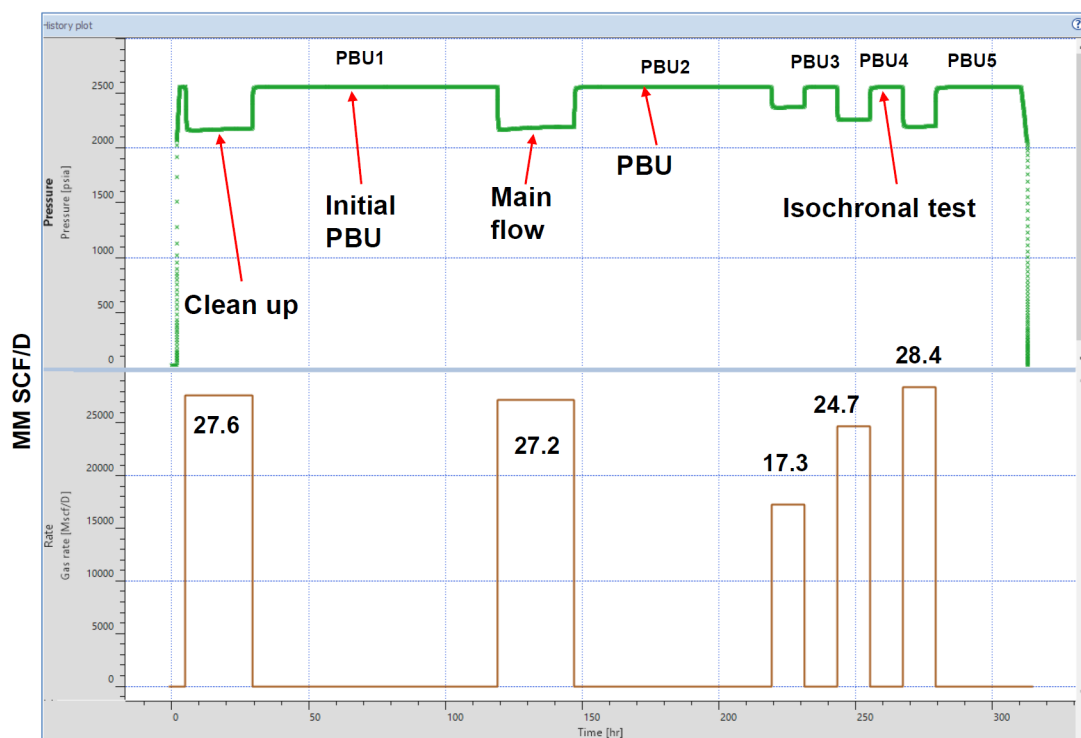


Figure 9—Pressure buildup and isochronal flow tests data in K3 reservoir

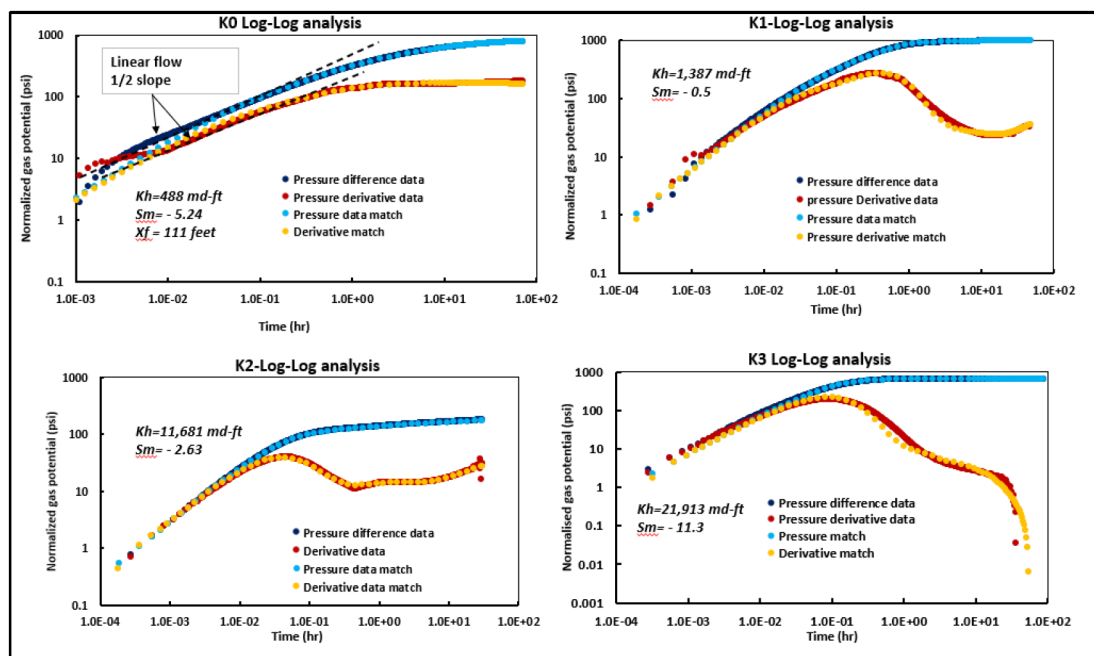


Figure 10—Log-Log Diagnosis and analysis of PBU data of K0, K1, K2 & K3 layers

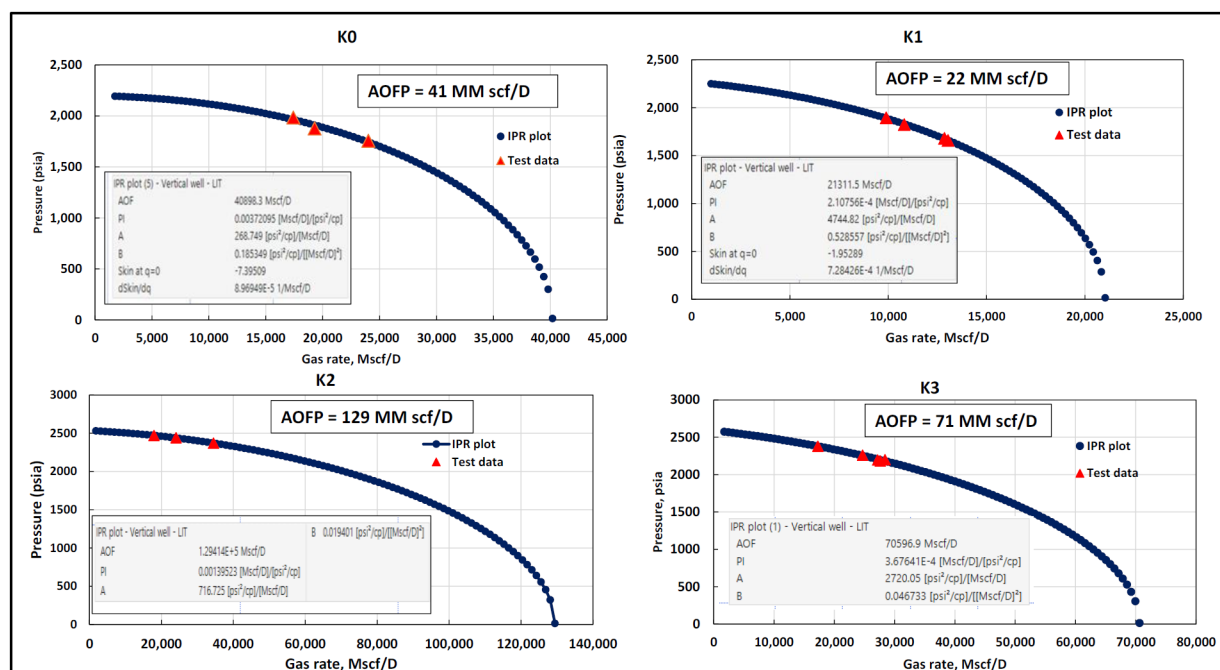


Figure 11—Matching of tests data of all layers and estimation of AOF from IPR plots based on LIT analysis using $m(p)$ function

The PTA results also suggest that the acid fracturing treatments effectively reduced near-wellbore damage and enhanced permeability, as evidenced by the improved pressure responses and flow test data. Table 2 summarizes the results of isochronal and pressure buildup tests.

Table 2—Isochronal Flow Tests and Pressure Buildup Results

Test	Linear flow	Kh, md-ft	Mech. Skin	Pavg, psia	AOF, MM scf/D	Rates Produced, MM scf/D
K0 PBU	Yes	488	-5.24	2,198	40.2	17.4 - 24
K1 PBU	No	1,387	-0.5	2,270	22	10.0 - 13
K2 PBU	No	11,681	-2.63	2,536	124-164	18 - 37
K3 PBU	No	21,913	-11.3	2,559	65-74	17 - 28.4

PVT and Core Analysis

Comprehensive PVT sampling and analysis were completed for all layers. Draft reports confirmed fluid properties suitable for dry gas production, with no anomalies detected. These data are critical for reservoir modeling and forecasting.

In parallel, a core analysis program was initiated, including conventional and special core analysis, rock mechanics testing, and petrography. These studies are expected to provide insights into fracture propagation behavior, proppant embedment risks, and mineralogical influences on acid-rock interactions.

Operational Observations

Zonal Isolation and Acid Placement. The use of diverting agents and stage-specific perforations ensured effective acid placement. No signs of crossflow were observed during testing, indicating successful zonal isolation.

Pressure Behavior. A consistent drop in treating pressure upon acid contact with the formation was observed in all stages, confirming acid-rock reaction and fracture initiation.

Post-Frac Monitoring. Plans were made to conduct production logging (SNL/PLT) to allocate flow contributions and monitor changes in zonal productivity over time.

Enhanced Productivity from Tight Intervals

The multi-stage acid fracturing approach applied to the Khuff reservoir demonstrated a significant uplift in gas productivity from tight dolomite intervals. Each of the four treated layers (K0 through K3) exhibited strong post-fracture performance, with K0, K2, and K3 showing excellent gas productivity, and K1 performing at a good level. This is a notable improvement over conventional completions, where tight layers typically contribute minimally or not at all.

Stable Production and Decline Behavior

Production testing revealed stable gas rates across all layers during the evaluation period, with no signs of early decline. The annual production decline rate for the multi-stage fracked well was observed at 7%, significantly lower than the 15–20% typical of conventionally completed Khuff wells. Figs. 12 & 13 show the decline performance of the fracked well and conventional well respectively indicating improved reservoir drainage and pressure support due to the stimulation.

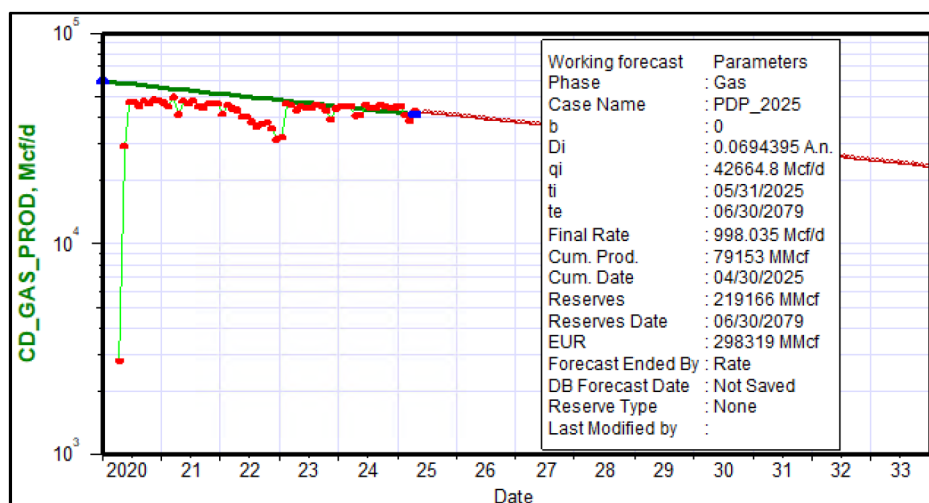


Figure 12—DCA analysis of fracked well showing 7% annual decline and estimation of 298 BCF EUR

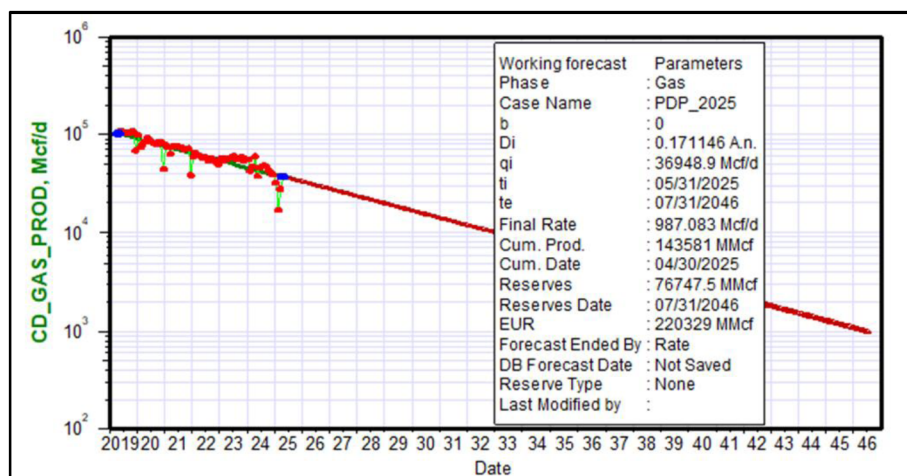


Figure 13—DCA analysis of conventional well showing 17% annual decline and estimation of 220 BCF EUR

Zonal Contribution and Logging Insights

Production logging (PLT) confirmed substantial contributions from the tight layers:

- K0 contributed up to 34–45% of total production, compared to ~10% in conventional wells.
- K3 showed a dramatic increase in contribution, from <1% in offset wells to 36–45% in the fracked well.
- K2 maintained a moderate contribution (~18–22%), while K1 remained low (~0–1%).

These results validate the effectiveness of the stimulation in unlocking reserves from previously underperforming zones. The well was surveyed for its production allocation 3 times during 2-years period with similar production percentage from each interval and showing slight improvement in K3 contribution in the latest survey. [Table-3](#) highlights the contribution from tight layers using the new stimulation approach as compared to nearby conventional wells. [Table-4](#) provides a time-lapse zonal contribution at different time indicating sustained contribution.

Table 3—PLT data suggest significant contribution from tight layers using new approach as compared to nearby conventional wells.

PLT Comparison		
Zone	Well fracked with new approach	Nearby well
K0	34%	4%
K1	1%	0%
K2	20%	95%
K3	45%	1%

Table 4—Zonal contribution for the new well at different time confirms sustained performance

Logging No.	Results			
	K0	K1	K2	K3
1	45%	0%	18%	37%
2	41%	1%	22%	36%
3	34%	1%	20%	45%

Reservoir Pressure Behavior

Post-frac pressure measurements indicated higher reservoir pressures than average field pressures in the stimulated intervals. The acid fracturing not only improved permeability and skin but also accessed compartments with better pressure support, contributing to improved productivity and production stability.

Discussion

Multi-stage acid fracturing has proven to be an effective technique for enhancing gas production and maximizing reserves in the tight dolomite intervals of the Khuff reservoir. These low- to moderate-porosity zones typically do not contribute meaningfully when perforated together with high-permeability layers. Under commingled production, contributions from these tight zones are limited—approximately 10% from K0 and less than 1% from K3. However, following multi-stage fracturing, individual layer tests demonstrated strong performance, as confirmed by isochronal testing and production logging surveys. Notably, the productivity of these tight layers post-fracturing was comparable to that of high-permeability zones. Pressure transient analysis (PTA) revealed negative mechanical skin values, indicating successful fracture stimulation.

Furthermore, the tight layers exhibited reservoir pressures higher than the average field pressure, highlighting their untapped reservoir potential. This is consistent with permeability-thickness (kh) values derived from pressure build-up (PBU) tests, isochronal testing, and absolute open flow potential (AOF) data. Production logging tool (PLT) surveys showed significant contributions from both the K0 and K3 layers corroborating PTA results. The well also demonstrated a reduced annual decline rate of 7%, compared to the typical 15–20% observed in conventional gas wells. The lower decline is attributed to the additional reserves accessed in the tight intervals. This revised completion strategy represents a major step forward in optimizing Khuff gas well performance by effectively targeting low- to moderate-porosity layers, unlocking previously untapped gas, and increasing the ultimate recoverable reserves (EUR).

Conclusion and Recommendation

This study underscores the importance of multi-stage acid fracturing to improve production and reserves in Khuff wells. Specifically, we reached the following conclusions:

1. The multi-stage acid fracturing approach successfully enhanced gas production from tight dolomite layers in the Khuff reservoir.
2. All the four stimulated zones contributed to production, with K0 and K3 demonstrating particularly strong performance.
3. The new completion strategy led to a markedly reduced production decline rate, indicating improved reservoir connectivity with the tight zones and supporting long-term deliverability and enhanced recovery.
4. Production logging, PTA and flow tests data are in good agreement and confirmed the effectiveness of the stimulation in unlocking reserves from untapped, low-permeability layers such as K0 and K3.

The key recommendations are

1. Adopt the multi-stage acid fracturing approach as a standard completion strategy for tight Khuff reservoir, particularly in wells targeting low- to moderate-porosity zones.
2. Continue the use of zonal isolation and staged stimulation to ensure effective acid placement and maximize reservoir contact.
3. Incorporate routine production logging to track zonal performance and optimize/improve future stimulation designs.
4. Evaluate the potential to extend this approach to other tight carbonate reservoirs with comparable characteristics to improve overall recovery efficiency and potentially reduce the well count needed to meet production targets.

Acknowledgement

The authors gratefully acknowledge the management of Bapco Upstream for their support and permission to publish this paper.

Nomenclature

- C = constant of Rawlins and Schellhardt gas equation, Mscf/D/psia²/cp
- kh = permeability-thickness, md-ft.
- m(p) = Pseudo-pressure function, psia²/cp
- n = exponent of Rawlins and Schellhardt equation.
- Xf = fracture half-length, feet.

Abbreviation

AOF	Absolute open flow potential.
BCF	Billion cubic feet.
DCA	Decline curve analysis.
EUR	Estimated ultimate recovery.
IPR	Inflow performance relationship.
MW	Molecular weight.
MSAG	Multi-stage acid fracturing.
PBU	Pressure buildup.
PLT	Production logging tool.
PTA	Pressure transient analysis.
SNL	Spectral noise log.

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